A High Speed 256-Bit Carry Look Ahead Adder Design Using 22nm Strained Silicon Technology

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***Abstract*—In this paper, a high speed 256-bit carry look ahead adder has been designed using 22nm strained silicon technology. The proposed adder combines the advantage of both the static and dynamic designs, which exhibits lower leakage, higher noise immunity and high speed. The speed performance of the proposed 256-bit adder is significantly improved by computing the even and the odd carries separately by using two separate Manchester carry chains. The circuit is simulated in HSPICE in the high performance 22nm PTM strained silicon CMOS technology with a supply voltage of V_{DD} = 0.8V. The simulation results show that the proposed 256-bit adder implemented using 8-bit adder modules shows significant operating speed improvement compared to the conventional 256-bit adder based on the standard 4-bit MCC adder modules.**

***Keywords*— Carry look ahead adders, Domino logic, Manchestor carry chain, noise immunity, technology scaling .**

I. INTRODUCTION

In digital computer design, adder is one of the most important component and it is used in various blocks of its architecture. In modern digital computers, addition is the most commonly used operation, since adders are not only used in arithmetic logic units[1], but also used for address generation in case of cache or memory access[2]. Addition is also a fundamental operation for many of the digital signal processing(DSP) or control systems. Hence improving the speed of binary adders will increase the efficiency of VLSI processors, DSP or control systems and digital computers. The classic high speed adder architectures include carry look ahead adders[3]-[7], carry skip adders[8], carry select adders[9]-[14], conditional sum adder[15], parallel prefix adders[16],[17], Kogge-Stone adder[17]

The power efficient VLSI circuits are of vital importance in electronic devices like cellular phones, smart phones, laptops etc, especially during this time of their increasing demands. The power consumption in scaled technologies can be reduced by two basic approaches: reducing the dynamic power consumption of the device when it is operated in active mode and leakage current reduction when it is operated in standby mode. To reduce the dynamic power consumption a lower supply voltage (V_{DD}) can be used.

The carry look ahead adder, used for addition of two binary numbers, is one of the fastest adders. Modified full adder circuits are used in carry look ahead adder for each bit position. The carry signals are generated independently for each group of k-bits using look ahead modules. Two carry signals are generated by the look ahead modules: carry generate and carry propagate. This reduces the delay time.

The Ripple carry adder consisting of N-bits can be implemented by connecting N full adders in series. The carry signal "ripple" from the least significant bit position to the most significant bit position and therefore it is called Ripple Carry Adder (RCA).

The architecture of carry select adder mainly consists of a multiplexer and dual ripple carry adders. In the addition process two assumptions are made: C_{in} = 0 and C_{in} = 1 and the carry signal is generated for both the cases. The role of the multiplexer is to select the correct sum and carry. The types of carry select adders include variable carry select adder and uniform carry select adder of which the first type reduces the delay in the C_{in} input.

In a Carry Skip Adder the words to be added are divided into groups, equal sized k-bits. In the group, carry propagates in ripple-carry manner. The delay of the carry skip adder is maximum when the carry signal generated is in the least significant bit position.

The conditional sum adder is another one of the fastest adders. In conditional sum adder, two bits can be added without waiting for the carry signal. In the addition process two assumptions are made: C_{in} = 0 and C_{in} = 1. The multiplexer selects the results sum_0, sum_1 and $carry_0$, $carry_1$ using the input carry signal as a multiplexer control.

To speed up the binary addition, a more flexible adder called the parallel prefix adder constructed from carry look ahead adder structure can be used, which involves three stages namely (1) Pre-processing stage which is used to generate and propagate signals (2) Carry generation network which is used to compute the carries and (3) a Post-processing stage which is used to compute the summation of input bits. Kogge-Stone adder is a modification of parallel prefix form of carry look ahead adder which provides the fastest addition operation.

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The paper is organized as follows. Section II describes the basic concepts and previous work. The architecture of proposed 256-bit Manchester carry chain adder and the simulation results are described in Section III and IV respectively. The conclusion is given in SectionV.

II. BASIC CONCEPTS AND PREVIOUS WORKS

Let $A=A_0\ A_1.....A_{n-2}\ A_{n-1}$ and $B= B_0\ B_1.....B_{n-2}\ B_{n-1}$ represents 2 n bit binary numbers to be added.

TABLE-1: Truth Table of Full Adder

A	B	$Carry_{in}$	SUM	CARRY	
0	0	0	0	0	
0	0	1	1	0	
0	1	0	1	0	
0	1	1	0	1	CARRY PROPO GATE
1	0	0	1	0	
1	0	1	0	1	
1	1	0	0	1	CARRY GENER ATE
1	1	1	1	1	

A. CARRY PROPAGATE

For four cases of the input combination as such (i)A=0,B=1,C_{in}=0 (ii)A=0,B=1,C_{in}=1 (iii)A=1,B=0,C_{in}=0 (iv)A=1,B=0,C_{in}=0. The input carry C_{in} is propagated to the output carry irrespective of both the inputs A&B and hence this process is known as carry propagation. The mathematical expression for carry propagation is obtained as follows.

The output carry is high(logic 1) for 2 cases of the input namely (i) A=0 & B=1 (ii) A=1 & B=0 ,these two cases resembles the function of an EXOR gate i.e when anyone of the inputs considered is high then the output is also high. Hence the carry propagation can be mathematically expressed as

$$C_p= A \oplus B \quad \text{----------(1)}$$

B. CARRY GENERATE

The mathematical expression for carry propagation is obtained as follows.

Here irrespective of the input carry the output carry is high for 2 cases of the inputs i.e A=1 & B=1 for both the cases. This resembles the function of an AND gate i.e when both the inputs considered are high then the carry output is also high. Hence the carry generation can be mathematically expressed as

$$C_G=A.B \quad \text{----------(2)}$$

In this Manchester Carry Chain Adder, domino logic is used to implement the carry propagate and generate circuits. the SUM circuits are implemented using static CMOS X-OR gates. The Manchester Carry Chain Adder has a provision to compute the odd and the even carries separately, this provision increases the speed of the adder considerably.

the equation for even carries:

$$h_0= g_0 + C_{-1} \quad \text{----------(3)}$$
$$h_2= G_2 + P_2G_0 \quad \text{----------(4)}$$
$$h_4= G_4 + P_4G_2 + P_4P_2G_0 \quad \text{----------(5)}$$
$$h_6= G_6 + P_6G_4 + P_6P_4G_2 + P_6P_4P_2G_0 \quad \text{----------(6)}$$
$$C_i= g_i + Z_i C_{i-1} \quad \text{----------(7)}$$

where $g_i=a_i.b_i$,

$Z_i= P_i= a_i \oplus b_i$

the sum bits of the conventional adder are given by

$$S_i= P_0 \oplus C_{i-1} \quad \text{----------(8)}$$

C is the carry input

$$C_1= G_i +Z_ig_{i-1} + Z_iZ_{i-1}g_{i-2}+.....+Z_iZ_{i-1}.....Z_ig_0+ Z_iZ_{i-1}Z_0C_{-1} \text{ ----(9)}$$

the equation for odd carriers are

$$h_1= G_1 + P_1C_{-1} \quad \text{----------(10)}$$
$$h_3= G3 +P_3G_1 +P_3P_1C_{-1} \quad \text{----------(11)}$$
$$h_5= G_5 +P_5G_3 +P_5P_3G_1 + P_5P_3P_1C_{-1} \quad \text{----------(12)}$$
$$h_7= G_7 + P_7G_5 + P_7P_5G_3 +P_7P_5P_3G_1 + P_7P_5P_3P_1C_{-1} \quad \text{---(13)}$$

where $G_{i=} g_i + g_{i-1}$ and $P_i = p_i.p_{i-1}.t_{i-2}$ & $t_i= a_i+b_i$ and $g_{-1}= C_{-1}$ & $t_{-1}=1$

the sum bits can be calculated as following

$$S_i= h_{i-1}.p_i + h_{i-1} (p_i \oplus t_{i-1}) \text{ for } i > 0 \quad \text{----------(14)}$$
$$S_0=p_0 \oplus C_{-1} \text{ for } i=0 \quad \text{----------(15)}$$

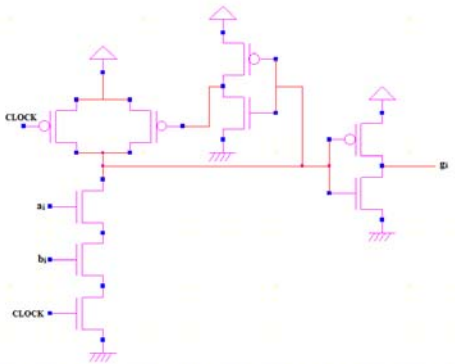


Figure 1. Implementation of g_i using Domino logic

The domino circuits for the computation of generate signals g_i and G_i are shown in fig. 1 and fig. 5 respectively, fig. 3 and fig. 4 shows the domino circuits for the computation of propagate signals p_i and P_i respectively, fig. 2 shows the domino circuit for the computation of t_i signal and the Manchester carry circuits for the computation of even and odd carries are shown in fig. 6 and fig. 7 respectively. The performance of the Manchester Carry Chain based Carry Look

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Ahead Adders can be further improved by providing carry skip facility.

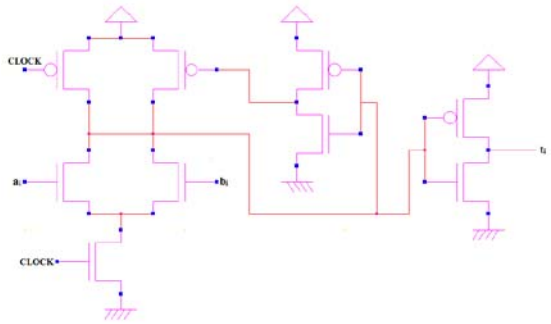


Figure 2. Implementation of t_i using domino logic

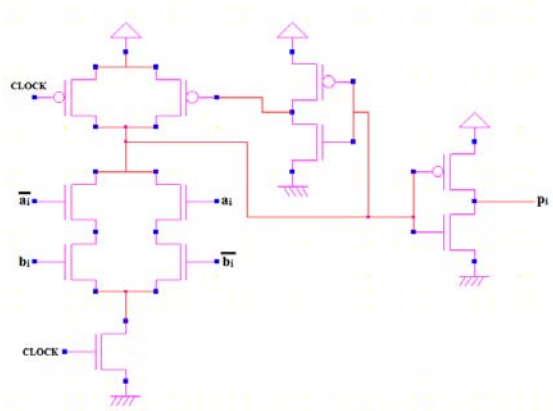


Figure 3.Implementation of p_i using domino logic

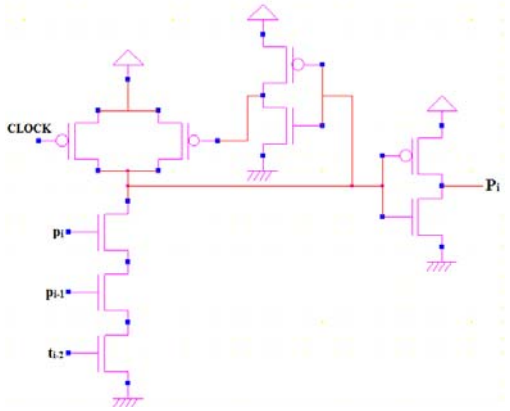


Figure 4.Implementation of P_i using domino logic

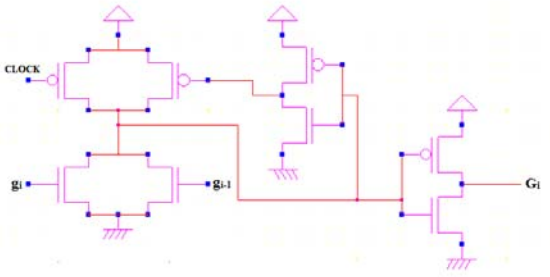


Figure 5.Implementation of G_i using domino logic

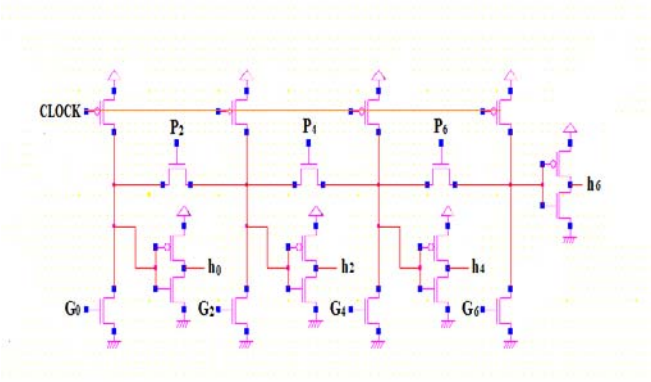


Figure 6. Implementation of MCC for computing EVEN carries

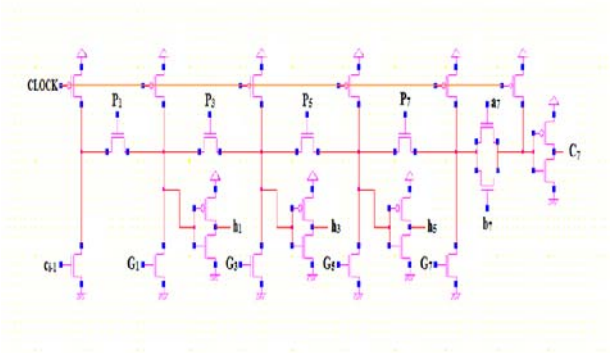


Figure 7. Implementation of MCC for computing ODD carries

III. ARCHETECTURE OF PROPOSED 256 Bit ADDER

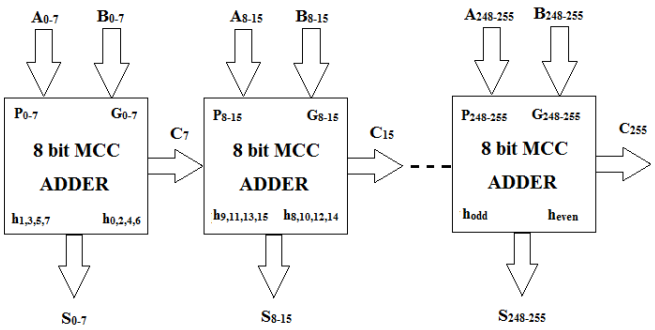


Figure 8. Architecture of 256-Bit CLA Adder

The conventional 16-, 32-, 64-, 128- and 256-Bit Manchester Carry Chain Adders are implemented by cascading two, four, eight, sixteen and thirty two 8-Bit Manchester Carry Chain Adder modules respectively. The 8-Bit Manchester Carry Chain Adder Module is implemented using 22nm strained silicon technology. Even though it is unfair to compare 22nm technology with 90nm technology, a comparison of the performance of these two technologies is given in the following table-2.

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TABLE-2: PERFORMANCE COMPARISON

ADDER LENGTH	DELAY(ps) (Conventional 90-nm Technology)	DELAY(ps) (PTM 22-nm Technology)
8-Bit	226.21	107.06
16-Bit	458.15	216.83
32-Bit	881.93	417.39
64-Bit	1718.20	813.17
128-Bit	-	1593.81
256-Bit	-	3139.81

The circuits have been implemented using 22nm high performance strained silicon technology which was introduced by PTM (Predictive Technology Model). The 22nm High Performance Strained Silicon Technology provides more than 50% improvement in the speed performance of 8-, 16-, 32-, 64-, 128-, 256-Bit adders.

IV. SIMULATION RESULTS

The simulations were performed using L=22nm technology along with the supply voltage V_{DD}=0.8V. The circuit has been simulated using HSPICE tool. Some of the inputs to the adder are shown in fig. 8 and the sum and the carry outputs are shown in the fig. 9 and fig. 10 respectively. Since the major part of the Carry Look Ahead Adder circuit has been implemented using domino logic, the circuit has two phases of operations, namely (i) Precharge phase and (ii) Evaluation phase. During the precharge phase the clock input is at low state and this low state helps the p-mos to turn ON, allowing the dynamic node to charge to V_{DD}. This charge at the dynamic node then passes through the inverter, which keeps the output state to be low, irrespective of the input states during the precharge phase. During the evaluation phase the clock input is made high and the logic operations of the circuits are carried out during this phase. The domino logic has a charge sharing problem and this short coming is overcome by employing a keeper architecture at the pull up tree.

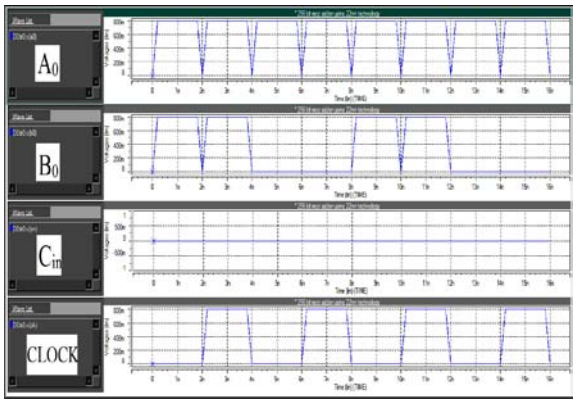


Figure 9. Inputs and Clock to the Adder

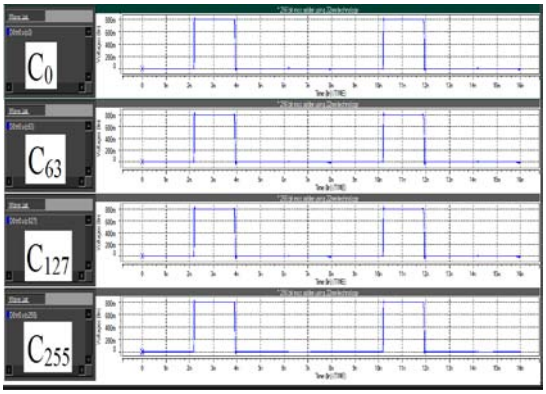


Figure 10. Carry output of the Adder

The simulations were performed at 30^0C and to analyse the effect of temperature on the circuit, the temperature was swept from 30^0C to 110^0C. The effect of temperature on the performance of the circuit is shown in fig. 11. The switching characteristic of the output signal with respect to variation in temperature is shown in figure 13. From figure 14, it is clear that in this 256-bit Carry Look Ahead Adder circuit, as the temperature increases from 30^0C to 110^0C, the delay has been changed from 3.846 nano seconds to 3.867 nano seconds. That is the delay is increased by 21 pico seconds. This can be explained with the help of following equations.

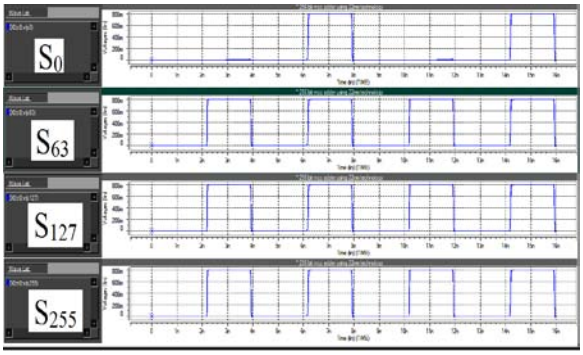


Figure 11. Sum output of the Adder

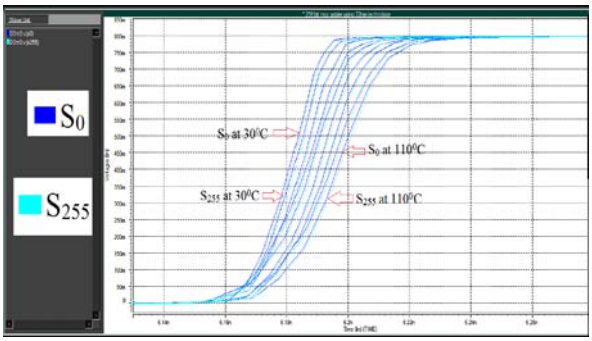


Figure 12. Temperature sweep of S_0 and S_{225}

The major part of the leakage current is the sub-threshold current which is given by

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$$I_{sub} = I_0 \, e^{\frac{(V_{GS} - V_{TH} - \gamma V_S + \eta V_{DS})}{n V_T}} \left(1 - e^{\frac{V_{DS}}{V_T}}\right) \quad (1)$$

Where $$I_0 = \mu_0 C_{OX} \frac{W}{L} (n-1) V_T^2 \quad (2)$$

where V_{GS} is the transistor gate to source voltage, V_{DS} is the transistor drain to source voltage, V_{TH} is the threshold voltage, V_T =KT/q is the thermal voltage, η is the DIBL coefficient, n is the sub-threshold swing coefficient of the transistor, μ_0 is the zero bias mobility, C_{OX} is the gate oxide capacitance and W, L are the width and length of the transistor respectively. The delay of domino logic circuits depends on the leakage current and the major part of leakage current is the sub-threshold current. Also the sub-threshold current in dynamic circuits is a function of operating temperature. The leakage current increases exponentially with respect to temperature and this increase in leakage current due to the increase in temperature will increase the circuit delay. The delay is measured between the rising edge of the clock signal to the rising edge of the output signal. That is the delay is the time interval between the clock and output signals crosses the mark $V_{DD}/2$. The power Vs time characteristics of 256-Bit Carry Look Ahead Adder is shown in fig. 15, and fig. 16 shows the power Vs temperature characteristics.

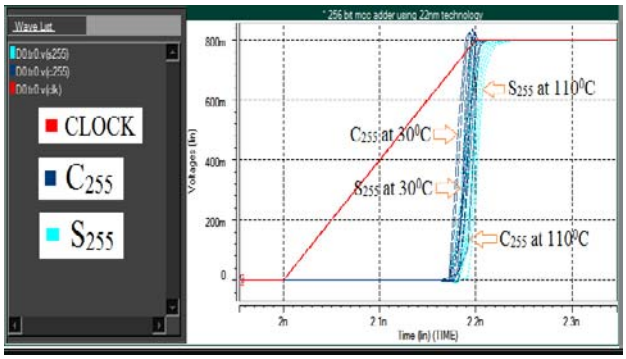


Figure 13. Temperature sweep of sum and carry output

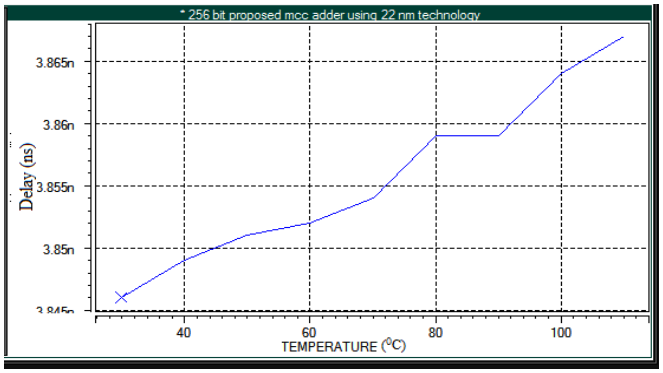


Figure 14. Delay vs Temperature characteristics

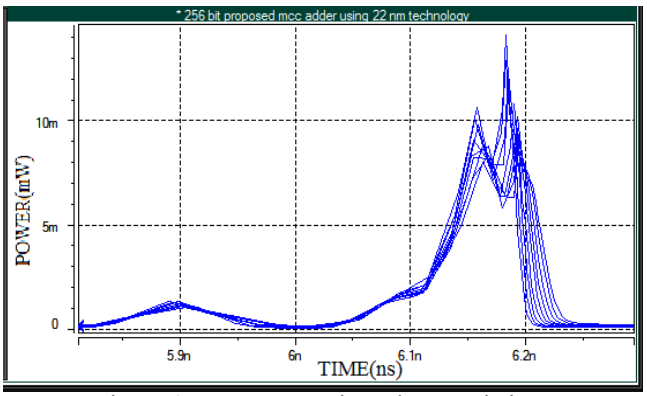


Figure 15. Power Vs Time characteristics

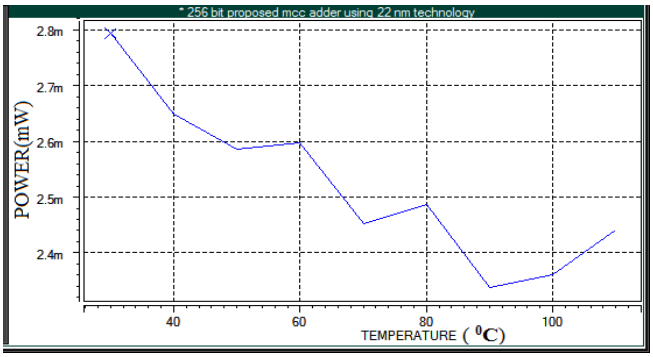


Figure 16. Power Vs Temperature characteristics

V. CONCLUSION

In this paper a high speed 256-bit Carry Look Ahead Adder, has been implemented and analysed using high performance 22nm strained silicon technology. The proposed adder provides more than 50% improvement in the speed performance of the Carry Look Ahead Adder. The delay of the 256-bit adder is found to be 3.846 nano seconds at 30^{0}C and 3.867 nano seconds at 110^{0}C. That is if the temperature is varied from 30^{0}C to 110^{0}C, the delay of the 256-bit adder will be increased by 21 pico seconds. As the temperature increases from 30^{0}C to 110^{0}C the power of the 256-Bit adder drops from 2.8 to 2.44 milli Watt. The supply voltage used in simulation is 0.8V. The simulation result shows that the circuit shows good noise immunity at 22nm technology.

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